

1. Let $f(x)$ be a function satisfying $f(x) + f(\pi - x) = \pi^2$, $\forall x \in \mathbb{R}$. Then $\int_0^\pi f(x) \sin x \, dx$ is equal to
[2023 (06 Apr Shift 2)]
 (1) $\frac{\pi^2}{4}$
 (2) $2\pi^2$
 (3) π^2
 (4) $\frac{\pi^2}{2}$
2. Let $f(x) = \frac{x}{(1+x^n)^{\frac{1}{n}}}$, $x \in \mathbb{R} - \{-1\}$, $n \in \mathbb{N}$, $n > 2$. If $f^n(x) = (f \circ f \circ f \dots \text{upto } n \text{ times})(x)$, then $\lim_{n \rightarrow \infty} \int_0^1 x^{n-2} (f^n(x)) dx$ is equal to
[2023 (06 Apr Shift 2)]
3. Let $[t]$ denote the greatest integer $\leq t$. Then $\frac{2}{\pi} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8[\operatorname{cosec} x] - 5[\cot x]) dx$ is equal to -----
[2023 (08 Apr Shift 1)]
4. Let $[t]$ denote the greatest integer function. If $\int_0^{2.4} [x^2] dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3} + \delta\sqrt{5}$, then $\alpha + \beta + \gamma + \delta$ is equal to
[2023 (08 Apr Shift 2)]
5. Let f be a continuous function satisfying $\int_0^{t^2} (f(x) + x^2) dx = \frac{4}{3}t^3$, $\forall t > 0$. Then $f\left(\frac{\pi^2}{4}\right)$ is equal to
[2023 (10 Apr Shift 2)]
 (1) $\pi^2 \left(1 - \frac{\pi^2}{16}\right)$
 (2) $-\pi \left(1 + \frac{\pi^3}{16}\right)$
 (3) $\pi \left(1 - \frac{\pi^3}{16}\right)$
 (4) $-\pi^2 \left(1 + \frac{\pi^2}{16}\right)$
6. The value of the integral $\int_{-\log_e 2}^{\log_e 2} e^x \left(\log_e \left(e^x + \sqrt{1 + e^{2x}} \right) \right) dx$ is equal to
[2023 (11 Apr Shift 1)]
 (1) $\log_e \left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} \right) - \frac{\sqrt{5}}{2}$
 (2) $\log_e \left(\frac{(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} \right) + \frac{\sqrt{5}}{2}$
 (3) $\log_e \left(\frac{2(2+\sqrt{5})}{\sqrt{1+\sqrt{5}}} \right) - \frac{\sqrt{5}}{2}$
 (4) $\log_e \left(\frac{\sqrt{2}(3-\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} \right) + \frac{\sqrt{5}}{2}$
7. For $m, n > 0$, let $\alpha(m, n) = \int_0^2 t^m (1 + 3t)^n dt$. If $11\alpha(10, 6) + 18\alpha(11, 5) = p(14)^6$, then p is equal to
[2023 (11 Apr Shift 1)]
8. Let the function $f: [0, 2] \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} e^{\min\{x^2, x - [x]\}}, & x \in [0, 1) \\ e^{[x - \log_e x]}, & x \in [1, 2] \end{cases}$, where $[t]$ denotes the greatest integer less than or equal to t . Then the value of the integral $\int_0^2 x f(x) dx$ is
[2023 (11 Apr Shift 2)]
 (1) $1 + \frac{3e}{2}$
 (2) $(e - 1) \left(e^2 + \frac{1}{2} \right)$
 (3) $2e - 1$
 (4) $2e - \frac{1}{2}$
9. If $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $\int_0^{\frac{\pi}{2}} f(\sin 2x) \sin x \, dx + \alpha \int_0^{\frac{\pi}{4}} f(\cos 2x) \cos x \, dx = 0$, then the value of α is
[2023 (11 Apr Shift 2)]
 (1) $\sqrt{2}$
 (2) $-\sqrt{3}$
 (3) $\sqrt{3}$
 (4) $-\sqrt{2}$
10. If $\int_{-0.15}^{0.15} |100x^2 - 1| dx = \frac{k}{3000}$, then k is equal to _____.
[2023 (12 Apr Shift 1)]

11. The value of $\frac{e^{-\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x} \tan^{50} x dx}{\int_0^{\frac{\pi}{4}} e^{-x} (\tan^{49} x + \tan^{51} x) dx}$ is
- [2023 (13 Apr Shift 2)]
- (1) 51
(2) 50
(3) 25
(4) 49
12. Let $f_n = \int_0^{\frac{\pi}{2}} \left(\sum_{k=1}^n \sin^{k-1} x \right) \left(\sum_{k=1}^n (2k-1) \sin^{k-1} x \right) \cos x dx$, $n \in \mathbb{N}$. Then $f_{21} - f_{20}$ is equal to
- [2023 (13 Apr Shift 2)]
13. If $\int_0^1 \frac{1}{(5+2x-2x^2)(1+e^{(2-4x)})} dx = \frac{1}{\alpha} \log_e \left(\frac{\alpha+1}{\beta} \right)$, $\alpha, \beta > 0$, then $\alpha^4 - \beta^4$ is equal to
- [2023 (15 Apr Shift 1)]
- (1) 19
(2) -21
(3) 0
(4) 21

ANSWER KEYS

1. (3) 2. (0) 3. (14) 4. (6) 5. (3) 6. (1) 7. (32) 8. (4)
9. (4) 10. (575) 11. (2) 12. (41) 13. (4)

1. (3)

Let

$$I = \int_0^\pi f(x) \sin x dx \quad \dots (1)$$

Now using the property $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$ we get,

$$\Rightarrow I = \int_0^\pi f(\pi-x) \sin(\pi-x) dx$$

$$\Rightarrow I = \int_0^\pi f(\pi-x) \sin x dx \quad \dots (2)$$

Adding (1) & (2), we get

$$2I = \int_0^\pi [f(x) + f(\pi-x)] \sin x dx$$

$$\Rightarrow 2I = \pi^2 \int_0^\pi \sin x dx \quad \{\text{as given } f(x) + f(\pi-x) = \pi^2\}$$

$$\Rightarrow 2I = \pi^2 [-\cos x]_0^\pi$$

$$\Rightarrow 2I = 2\pi^2$$

$$\Rightarrow I = \pi^2$$

Hence this is the correct option.

2. (0)

Given,

$$f(x) = \frac{x}{(1+x^n)^{\frac{1}{n}}}, \quad x \in \mathbb{R} - \{-1\}, \quad n \in \mathbb{N}, \quad n > 2,$$

$$f^n(x) = (f \circ f \circ f \dots \text{upto } n \text{ times}) (x),$$

$$\text{Now finding } f(f(x)) = \frac{f(x)}{\left[1 + \left(f(x)\right)^{\frac{1}{n}}\right]^{\frac{1}{n}}}$$

$$\Rightarrow f(f(x)) = \frac{\frac{x}{(1+x^n)^{\frac{1}{n}}}}{\left[1 + \left(\frac{x}{(1+x^n)^{\frac{1}{n}}}\right)^{\frac{1}{n}}\right]^{\frac{1}{n}}}$$

$$\Rightarrow f(f(x)) = \frac{x}{\left[1 + \left(\frac{x}{(1+x^n)^{\frac{1}{n}}}\right)^{\frac{1}{n}}\right]^{\frac{1}{n}}}$$

$$\Rightarrow f(f(x)) = \frac{x}{\left[1 + 2x^n\right]^{\frac{1}{n}}}$$

$$\text{Similarly, } f^n(x) = \frac{x}{(1+nx^n)^{\frac{1}{n}}}$$

Now solving the integral we get,

$$I = \int_0^1 \frac{x^{n-1}}{(1+nx^n)^{\frac{1}{n}}} dx$$

$$\text{Now let, } 1 + nx^n = t^n$$

$$\Rightarrow n^2 x^{n-1} dx = nt^{n-1} dt$$

$$I = \int_1^{(1+n)^{\frac{1}{n}}} \frac{1}{t} \cdot \frac{1}{n} dt$$

$$\Rightarrow I = \left[\frac{1}{n} \ln t \right]_1^{(1+n)^{\frac{1}{n}}}$$

$$\Rightarrow I = \frac{1}{n(n-1)} \left((1+n)^{1-\frac{1}{n}} - 1 \right)$$

Now solving the limit we get,

$$\lim_{n \rightarrow \infty} \frac{(1+n)^{1-\frac{1}{n}} - 1}{n(n-1)}$$

Now let $n = \frac{1}{h}$ so the limit changes to,

$$\lim_{h \rightarrow 0} \frac{\left(1 + \frac{1}{h}\right)^{1-h} - 1}{\frac{1}{h} \left(\frac{1}{h} - 1\right)}$$

$$= \lim_{h \rightarrow 0} \frac{\left(1 + \frac{1}{h}\right)^{1-h} - 1}{\frac{1}{h} \left(\frac{1}{h} - 1\right)}$$

$$= \lim_{h \rightarrow 0} \frac{(1-h)}{\left(\frac{1}{h} - 1\right)}$$

$$= \lim_{h \rightarrow 0} \frac{h(1-h)}{(1-h)}$$

$$= \lim_{h \rightarrow 0} h = 0$$

3. (14)

Let,

$$I = \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8[\operatorname{cosec} x] - 5[\cot x]) dx$$

$$\Rightarrow I = 8 \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} [\operatorname{cosec} x] dx - 5 \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} [\cot x] dx$$

$$\Rightarrow I = 8I_1 - 5I_2, \text{ where } I_1 = 8 \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} [\operatorname{cosec} x] dx \text{ \& } I_2 = 5 \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} [\cot x] dx$$

Now solving,

$$I_1 = \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} [\operatorname{cosec} x] dx$$

$$\Rightarrow I_1 = \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 1 dx = \frac{2\pi}{3}$$

As when $x \in \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$, $\operatorname{cosec} x \in [1, 2)$,

So, $[\operatorname{cosec} x] = 1$

Now solving,

$$I_2 = \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} [\cot x] dx$$

$$\Rightarrow I_2 = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 1 dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 0 dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} (-1) dx + \int_{\frac{3\pi}{4}}^{\frac{5\pi}{6}} (-2) dx$$

$$\Rightarrow I_2 = \left(\frac{\pi}{4} - \frac{\pi}{6}\right) - \left(\frac{3\pi}{4} - \frac{\pi}{2}\right) - 2\left(\frac{5\pi}{6} - \frac{3\pi}{4}\right)$$

$$\Rightarrow I_2 = -\frac{\pi}{3}$$

Required value will be,

$$\frac{2}{\pi} I = \frac{2}{\pi} \left[8 \times \frac{2\pi}{3} + 5 \times \frac{\pi}{3} \right] = 14$$

4. (6)

Given that $\int_0^{2.4} [x^2] dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3} + \delta\sqrt{5}$, then $\alpha + \beta + \gamma + \delta$

$$\text{We know that for } 1 \leq x < 2, [x] = 1 \Rightarrow \int_0^{2.4} [x^2] dx = \int_0^1 0 dx + \int_1^{\sqrt{2}} 1 dx + \int_{\sqrt{2}}^{\sqrt{3}} 2 dx + \int_{\sqrt{3}}^2 3 dx + \int_2^{\sqrt{5}} 4 dx + \int_{\sqrt{5}}^{2.4} 5 dx$$

$$= 0 + (\sqrt{2} - 1) + 2(\sqrt{3} - \sqrt{2}) + 3(2 - \sqrt{3}) + 4(\sqrt{5} - 2) + 5(2.4 - \sqrt{5})$$

$$= 9 - \sqrt{2} - \sqrt{3} - \sqrt{5}$$

This is in the form of $\alpha + \beta\sqrt{2} + \gamma\sqrt{3} + \delta\sqrt{5}$

$$\alpha = 9, \beta = -1, \gamma = -1, \delta = -1$$

$$\Rightarrow \alpha + \beta + \gamma + \delta = 9 - 1 - 1 - 1 = 6$$

Hence this is the required answer.

5. (3)

Given equation is $\int_0^{t^2} (f(x) + x^2) dx = \frac{4}{3} t^3, \forall t > 0$

According to Newton Leibnitz theorem we have $\frac{d}{dx} \left(\int_{u(x)}^{v(x)} f(t) dt \right) = f(v(x)) \times v'(x) - f(u(x)) \times u'(x)$

Apply Newtons Leibnitz theorem in the given equation.

$$\Rightarrow (f(t^2) + t^4) 2t - 0 = 4t^2$$

$$\Rightarrow f(t^2) + t^4 = 2t$$

$$\Rightarrow f(x^2) = -x^4 + 2x$$

$$\Rightarrow f(x) = -x^2 + 2\sqrt{x}$$

$$\Rightarrow f\left(\frac{\pi^2}{4}\right) = -\frac{\pi^4}{4^2} + 2 \times \frac{\pi}{2}$$

$$= \frac{-\pi^4}{16} + \pi$$

$$= \pi \left(1 - \frac{\pi^3}{16} \right)$$

Hence this is the correct option.

6. (1)

$$\text{Let } I = \int_{-\log_e 2}^{\log_e 2} e^x \left[\log_e (e^x + \sqrt{1 + e^{2x}}) \right] dx$$

Let us substitute $e^x = t$

$$\Rightarrow e^x dx = dt$$

$$\text{Lower limit} = e^{-\log_e 2} = \frac{1}{2}$$

$$\text{Upper limit} = e^{\log_e 2} = 2$$

$$\Rightarrow I = \int_{\frac{1}{2}}^2 \frac{1}{t} \times \log_e [t + \sqrt{1 + t^2}] dt$$

Apply integration by-parts.

$$\int u(t)v(t)dt = u(t) \int v(t)dt - \int (u'(t) \int v(t)dt) dt$$

$$\text{Take } u(t) = \log_e [t + \sqrt{1 + t^2}], v(t) = 1$$

$$\Rightarrow u'(t) = \frac{1}{t + \sqrt{1 + t^2}} \left(1 + \frac{2t}{2\sqrt{1 + t^2}} \right) = t$$

$$= \left[t \ln (\sqrt{t^2 + 1} + t) \right]_{\frac{1}{2}}^2 - \int_{\frac{1}{2}}^2 \frac{t}{\sqrt{t^2 + 1}} dt$$

$$= \left[t \ln (\sqrt{t^2 + 1} + t) - \sqrt{t^2 + 1} \right]_{\frac{1}{2}}^2$$

$$= \left[2 \ln (\sqrt{5} + 2) - \sqrt{5} \right] - \left[\frac{1}{2} \ln \left(\sqrt{\frac{5}{4} + \frac{1}{4}} + \frac{1}{2} \right) - \sqrt{\frac{5}{4}} \right]$$

$$= \ln \left(\frac{\sqrt{2}(2 + \sqrt{5})^2}{\sqrt{1 + \sqrt{5}}} \right) - \frac{\sqrt{5}}{2}$$

Hence this is the correct option.

7. (32)

Given,

$$\alpha(m, n) = \int_0^2 t^m (1 + 3t)^n dt$$

Now using integration by parts we get,

$$\Rightarrow \alpha(m, n) = (1 + 3t)^n \cdot \frac{t^{m+1}}{m+1} \Big|_0^2 - \int_0^2 n(1 + 3t)^{n-1} \times 3 \cdot \frac{t^{m+1}}{m+1} dt$$

$$\Rightarrow (m+1)\alpha(m, n) = (1 + 3t)^n \cdot (t^{m+1}) \Big|_0^2 - 3n(m+1) \int_0^2 (1 + 3t)^{n-1} \times t^{m+1} dt$$

$$\Rightarrow (m+1)\alpha(m, n) = (1 + 3 \times 2)^n \cdot (2^{m+1}) - 3n(m+1)\alpha(m+1, n-1)$$

$$\Rightarrow (m+1)\alpha(m, n) = 7^n \cdot 2^{m+1} - 3n(m+1)\alpha(m+1, n-1)$$

$$\Rightarrow (m+1)\alpha(m, n) + 3n\alpha(m+1, n-1) = 7^n \cdot 2^{m+1}$$

Now put $m = 10, n = 6$ in above equation we get,

$$11\alpha(10, 6) + 18\alpha(11, 5) = 7^6 \cdot 2^{11} = 32 \times (14)^6$$

Hence, on comparing with $11\alpha(10, 6) + 18\alpha(11, 5) = p \times (14)^6$ we get,

$$\Rightarrow p = 32$$

8. (4)

Given,

$$f(x) = \begin{cases} e^{\min\{x^2, x - [x]\}}, & x \in [0, 1) \\ e^{[x - \log_e x]}, & x \in [1, 2] \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} e^{x^2} & x \in [0, 1) \\ e, & x \in [1, 2] \end{cases}$$

As $x - \ln x \in [1, 2)$ for $x \in [1, 2]$, so $[x - \ln x] = 1$ and $x - [x] = \{x\}$, so $x \in (0, 1), x^2 < \{x\}$

Now solving the integral we get, $\int_0^2 x f(x) dx = \int_0^1 x \cdot e^{x^2} dx + \int_1^2 x \cdot e dx$

Now in first integral, let $x^2 = t \Rightarrow 2x dx = dt$ we get,

$$\Rightarrow \int_0^2 x f(x) dx = \left[\frac{1}{2} \int_0^1 e^t dt \right] + \left[e \cdot \frac{x^2}{2} \right]_1^2$$

$$\Rightarrow \int_0^2 x f(x) dx = \left(\frac{e-1}{2} \right) + \frac{3e}{2}$$

$$\Rightarrow \int_0^2 x f(x) dx = 2e - \frac{1}{2}$$

9. (4)

Given,

$$\int_0^{\frac{\pi}{2}} f(\sin 2x) \sin x \, dx + \alpha \int_0^{\frac{\pi}{4}} f(\cos 2x) \cos x \, dx = 0$$

$$\text{Now, let } I = \int_0^{\frac{\pi}{2}} f(\sin 2x) \cdot \sin x \, dx$$

Now using property $\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$ we get,

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} f(\sin 2x) \sin x \, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(\sin 2x) \cdot \sin x \, dx$$

Now using property $\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx$ we get,

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} f(\cos 2x) \sin \left(\frac{\pi}{4} - x \right) dx + \int_0^{\frac{\pi}{4}} f \left(\sin 2 \left(\frac{\pi}{4} + x \right) \right) \sin \left(\frac{\pi}{4} + x \right) dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} f(\cos 2x) \left(\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x \right) dx + \int_0^{\frac{\pi}{4}} f(\cos 2x) \left(\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x \right) dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} f(\cos 2x) (\sqrt{2} \cos x) dx$$

So, the putting the value of I in $\int_0^{\frac{\pi}{2}} f(\sin 2x) \sin x \, dx + \alpha \int_0^{\frac{\pi}{4}} f(\cos 2x) \cos x \, dx = 0$ we get, $\alpha = -\sqrt{2}$

10. (575)

Given,

$$\int_{-0.15}^{0.15} |100x^2 - 1| \, dx = \frac{k}{3000}$$

Now we know that, $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$ if $f(x)$ is even

$$\text{So, } \int_{-0.15}^{0.15} |100x^2 - 1| \, dx = 2 \int_0^{0.15} |100x^2 - 1| \, dx$$

$$\text{Now solving, } 2 \int_0^{0.15} |100x^2 - 1| \, dx = 2 \left(\int_0^{0.1} -(100x^2 - 1) \, dx + \int_{0.1}^{0.15} 100x^2 - 1 \, dx \right)$$

$$\Rightarrow 2 \int_0^{0.15} |100x^2 - 1| \, dx = 2 \left(\left[x - \frac{100x^3}{3} \right]_0^{0.1} + \left[\frac{100x^3}{3} - x \right]_{0.1}^{0.15} \right)$$

$$\Rightarrow 2 \int_0^{0.15} |100x^2 - 1| \, dx = 2 \left(\left(\frac{1}{10} - \frac{1}{30} \right) + \left(\frac{100 \times 15^3}{3 \times 100^3} - \frac{15}{100} \right) - \left(\frac{1}{30} - \frac{1}{10} \right) \right)$$

$$\Rightarrow 2 \int_0^{0.15} |100x^2 - 1| \, dx = 2 \left(\frac{2}{30} + \left(\frac{225}{2000} - \frac{15}{100} \right) + \frac{2}{30} \right)$$

$$\Rightarrow 2 \int_0^{0.15} |100x^2 - 1| \, dx = 2 \left(\frac{4}{30} + \left(\frac{-3}{80} \right) \right) = \left(\frac{8}{30} - \frac{3}{40} \right) = \frac{32-9}{120} = \frac{23}{120} = \frac{575}{3000}$$

$$\Rightarrow 2 \int_0^{0.15} |100x^2 - 1| \, dx = \frac{575}{3000}$$

Now comparing with $\int_{-0.15}^{0.15} |100x^2 - 1| \, dx = \frac{k}{3000}$, we get $k = 575$

11. (2)

Let,

$$I = \frac{e^{-\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x} \tan^{50} x \, dx}{\int_0^{\frac{\pi}{4}} e^{-x} (\tan^{49} x + \tan^{51} x) \, dx}$$

$$\text{Now let } I_1 = \int_0^{\frac{\pi}{4}} e^{-x} (\tan^{49} x + \tan^{51} x) \, dx$$

Now solving, $I_2 = \int_0^{\frac{\pi}{4}} e^{-x} \tan^{50} x \, dx$ using byparts we get,

$$I_2 = \int_0^{\frac{\pi}{4}} e^{-x} \tan^{50} x \, dx$$

$$\Rightarrow I_2 = \left[-e^{-x} \tan^{50} x \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} 50 \tan^{49} x \sec^2 x (-e^{-x}) \, dx$$

$$\Rightarrow I_2 = -e^{-\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} 50 \tan^{49} x (1 + \tan^2 x) e^{-x} \, dx$$

$$\Rightarrow I_2 = -e^{-\frac{\pi}{4}} + 50 \int_0^{\frac{\pi}{4}} (\tan^{49} x + \tan^{51} x) e^{-x} \, dx$$

$$\Rightarrow I_2 = -e^{-\frac{\pi}{4}} + 50 I_1$$

$$\Rightarrow \int_0^{\frac{\pi}{4}} e^{-x} \tan^{50} x \, dx + e^{-\frac{\pi}{4}} = 50 I_1$$

Now putting the given integral I we get,

$$I = \frac{e^{-\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x} \tan^{50} x \, dx}{\int_0^{\frac{\pi}{4}} e^{-x} (\tan^{49} x + \tan^{51} x) \, dx}$$

$$\Rightarrow I = \frac{50 I_1}{I_1} = 50$$

12. (41)

Given,

$$f_n = \int_0^{\pi/2} \left(\sum_{k=1}^n \sin^{k-1} x \right) \left(\sum_{k=1}^n (2k-1) \sin^{k-1} x \right) \cos x dx$$

Now let, $\sin x = t$

$$\Rightarrow \cos x dx = dt$$

$$\text{So, } f_n = \int_0^1 \left(\sum_{k=1}^n t^{k-1} \right) \left(\sum_{k=1}^n (2k-1) t^{k-1} \right) dt$$

$$\Rightarrow f_n = \int_0^1 (1+t+t^2+\dots+t^{n-1}) (1+3t+5t^2+\dots+(2n-1)t^{n-1}) dt$$

Now multiply and divide by $\sqrt{t}$ we get,

$$\Rightarrow f_n = \int_0^1 \frac{(t^{\frac{1}{2}} + t^{\frac{3}{2}} + t^{\frac{5}{2}} + \dots + t^{\frac{2n-1}{2}})}{\sqrt{t}} (1+3t+5t^2+\dots+(2n-1)t^{n-1}) dt$$

$$\Rightarrow f_n = \int_0^1 \left(t^{\frac{1}{2}} + t^{\frac{3}{2}} + t^{\frac{5}{2}} + \dots + t^{\frac{2n-1}{2}} \right) \left(t^{\frac{-1}{2}} + 3t^{\frac{1}{2}} + 5t^{\frac{3}{2}} + \dots + (2n-1)t^{\frac{2n-3}{2}} \right) dt$$

Now let $t^{\frac{1}{2}} + t^{\frac{3}{2}} + t^{\frac{5}{2}} + \dots + t^{\frac{2n-1}{2}} = z$

$$\Rightarrow \frac{1}{2} \left(t^{\frac{-1}{2}} + 3t^{\frac{1}{2}} + 5t^{\frac{3}{2}} + \dots + (2n-1)t^{\frac{2n-3}{2}} \right) dt = dz$$

Hence, the integral becomes,

$$\Rightarrow f_n = 2 \int_0^n z dz$$

$$\Rightarrow f_n = [z^2]_0^n = n^2$$

$$\text{Hence, } f_{21} - f_{20} = 21^2 - 20^2 = 41$$

13. (4)

Given,

$$\int_0^1 \frac{1}{(5+2x-2x^2)(1+e^{(2-4x)})} dx = \frac{1}{\alpha} \log_e \left(\frac{\alpha+1}{\beta} \right)$$

Now let,

$$I = \frac{1}{2} \int_0^1 \frac{1}{\left(\left(\frac{5}{2} - (x^2 - x) \right) \right) \left(1 + e^{-4\left(x - \frac{1}{2}\right)} \right)} dx$$

$$\Rightarrow I = \frac{1}{2} \int_0^1 \frac{1}{\left(\left(\frac{11}{4} - \left(x - \frac{1}{2}\right)^2 \right) \right) \left(1 + e^{-4\left(x - \frac{1}{2}\right)} \right)} dx$$

Now let $x - \frac{1}{2} = t \Rightarrow dx = dt$

So, integral becomes,

$$I = \frac{1}{2} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right) (1 + e^{-4t})} dt$$

$$\Rightarrow I = \frac{1}{2} \int_{-\frac{1}{2}}^0 \frac{1}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right) (1 + e^{-4t})} dt + \underbrace{\frac{1}{2} \int_0^{\frac{1}{2}} \frac{1}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right) (1 + e^{-4t})} dt}_{I_2}$$

$$\text{Now solving } I_2 = \frac{1}{2} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right) (1 + e^{-4t})} dt,$$

Let $t = -z$, we get $dt = -dz$

$$\text{So, } I_2 = -\frac{1}{2} \int_{\frac{1}{2}}^0 \frac{dz}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - z^2 \right) (1 + e^{4z})}$$

$$\Rightarrow I_2 = \frac{1}{2} \int_0^{\frac{1}{2}} \frac{dt}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right) (1 + e^{4t})}$$

Now putting the value of I_2 in I we get,

$$I = \frac{1}{2} \int_0^{\frac{1}{2}} \frac{1}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right) (1 + e^{-4t})} + \frac{1}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right) (1 + e^{4t})} dt$$

$$\Rightarrow I = \frac{1}{2} \int_0^{\frac{1}{2}} \frac{e^{4t}}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right) (e^{4t} + 1)} + \frac{1}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right) (1 + e^{4t})} dt$$

$$\Rightarrow I = \frac{1}{2} \int_0^{\frac{1}{2}} \frac{1}{\left(\left(\frac{\sqrt{11}}{2} \right)^2 - t^2 \right)} dt$$

$$\Rightarrow I = \frac{1}{2} \times \frac{1}{2 \left(\frac{\sqrt{11}}{2} \right)} \left[\ln \left| \frac{\frac{\sqrt{11}}{2} + t}{\frac{\sqrt{11}}{2} - t} \right| \right]_0^{\frac{1}{2}}$$

$$\Rightarrow I = \frac{1}{2\sqrt{11}} \ln \left(\frac{\sqrt{11}+1}{\sqrt{11}-1} \right) = \frac{1}{2\sqrt{11}} \ln \left(\frac{(\sqrt{11}+1)^2}{10} \right)$$

$$\Rightarrow I = \frac{1}{\sqrt{11}} \ln \left(\frac{\sqrt{11}+1}{\sqrt{10}} \right)$$

Now on comparing with $\frac{1}{\alpha} \log_e \left(\frac{\alpha+1}{\beta} \right)$ we get,

$$\Rightarrow \alpha = \sqrt{11}, \beta = \sqrt{10} \Rightarrow \alpha^4 - \beta^4 = 21$$